



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

Project Task 4

(Making The Connections – LAN and WAN)

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Section : 02

Group : 3 – NotAvailable

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4.1 Identify the work areas on floor plan

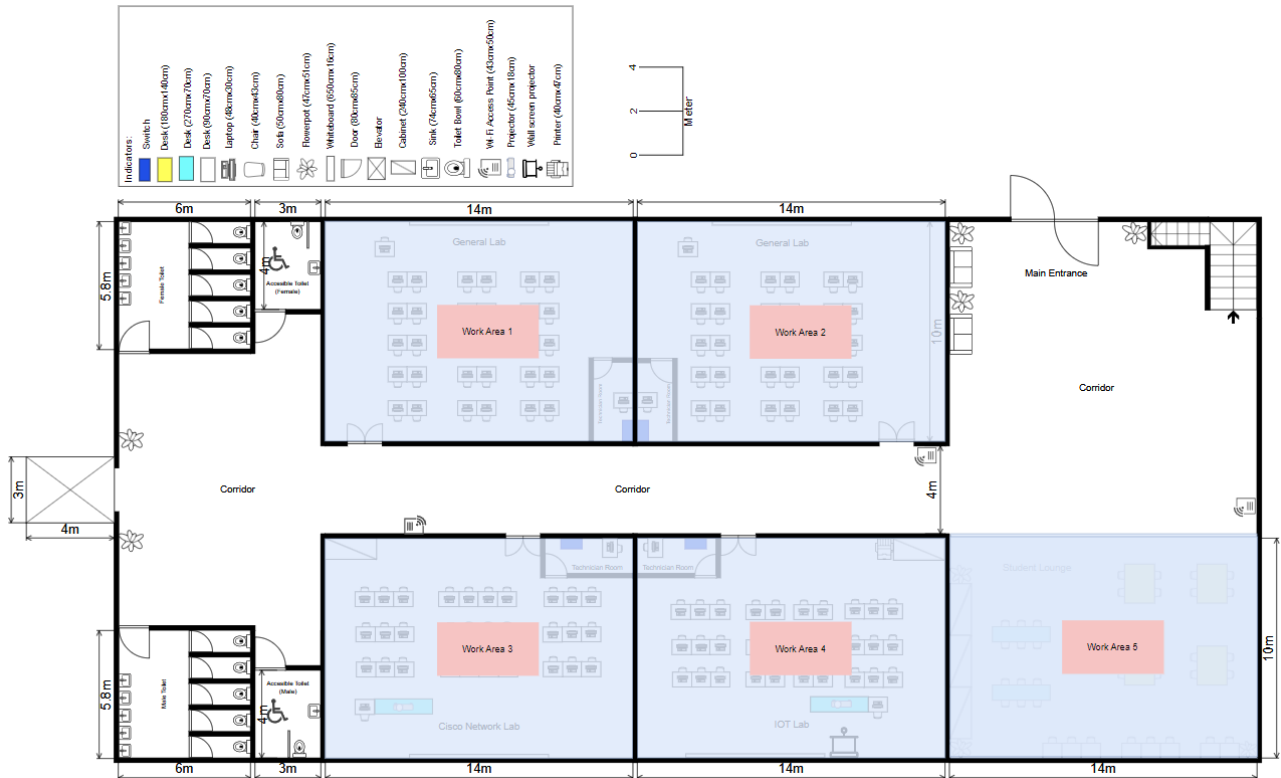


Figure 4.1.1: Work Area at Ground Floor

Work Area 1 and 2, as shown in Figure 4.1.1, are both general labs spanning 14 meters in length and 10 meters in width. These general labs are intended for student use during classes. Each lab is equipped with a total of 64 PC workstations, including those in each technician room, with 31 PC workstations in each general lab. Consequently, the general labs require three 24-port switches to connect all the workstations.

Next, for the Work Area 3 which is located at the bottom side of the ground floor as shown in Figure 4.1.1, it is a Cisco Network Lab where both lecturers and students can engage in various activities to facilitate learning and practical skill development. There are 30 workstations for students, one workstation and projector for lecturers to work with their teaching materials and one workstation located in the technician room to make technician staffs' work easier. In addition to that, one wireless access point will be installed and placed on a corridor outside of the network lab.

Work Area 4 shown in Figure 4.1.1 is an IOT lab which consists of 32 workstations. This dynamic space facilitates hands-on learning, research, and development in the IoT field. Additionally, it features a technician room, serving as a crucial support hub for maintenance, troubleshooting, and technical assistance.

Work Area 5 shown in Figure 4.1.1 is a student lounge which is a vibrant and welcoming space, spanning 14 meters in length and 10 meters in width, creating an expansive area for students to relax and engage in various activities. The design of the lounge is carefully curated to foster a comfortable and inclusive atmosphere. There are no workstations provided but students can bring their own devices and internet connection will be available as there is a wireless access point nearby.

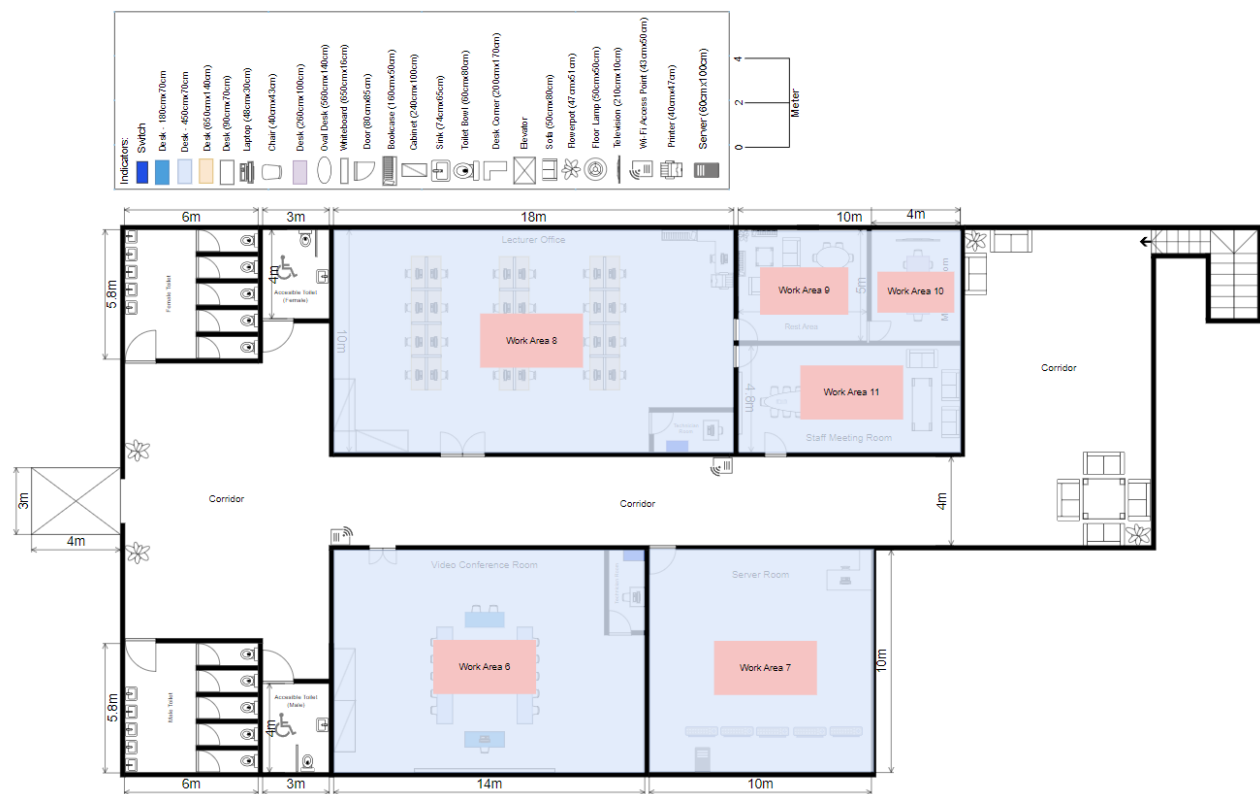


Figure 4.1.2: Work Area at First Floor

Work Area 6, depicted in Figure 4.1.2, serves the dual purpose of a video conference room and a server room. The video conference room is thoughtfully designed with a streamlined approach, featuring a central high-quality projector, a dedicated PC, and a strategically positioned screen for

optimal visibility. This setup ensures the room is conducive to effective presentations, collaborative work, and virtual meetings.

Meanwhile, Work Area 7, the server room is specifically designed to provide a dedicated space for the storage and protection of crucial data, along with the implementation of necessary monitoring and security measures. The server room houses a PC workstation in addition to four servers. Consequently, a switch is required to connect all PC workstations, the projector, and the four servers.

Moving to the upper side of the floorplan shown in Figure 4.1.2, Work Area 8 is a lecturer office which is available to all lecturers for them to prepare lecture materials, conduct academic research, grade students' assignments, exams, and other assessments and others. There are 24 workstations located in the middle of the office for lecturers, one workstation placed at the right corner and beside the printer to increase convenience, connectivity, and workflow efficiency. When entering the lecturer's office, there is a technician room which consists of one workstation and two 24-port switches. One wireless point is also located outside of the office.

Work Area 9 and Work Area 10 shown in Figure 4.1.2 are the rest area and meeting room. The rest area provides a comfortable and relaxing environment, designed to offer a brief respite from work-related activities. It offers a welcoming atmosphere where individuals can unwind, recharge, and socialize. Then, the meeting room is equipped with a huge television, making it an ideal setting for productive gatherings and discussions. This multifunctional space caters to the diverse needs of those using it, ensuring a harmonious blend of relaxation and productivity within its confines.

Work Area 11 shown in Figure 4.1.2 is a staff meeting room which aims to facilitate efficient and collaborative discussions among team members. There are essential tools to aid in presentations and brainstorming sessions within this space. Prominently featured is a whiteboard, which allows for visualizing ideas and concepts during meetings. Besides, the projector enhances the room's capability to display presentations and share information effectively. The room layout includes a central table, providing a focal point for discussions, while the inclusion of comfortable sofas ensures that participants can engage in these meetings in a relaxed and conducive environment.

This well-appointed staff meeting room is designed to encourage productivity, communication, and the exchange of ideas among the team.

4.2 The distribution and connection of devices

4.2.1 Network Distribution Diagram

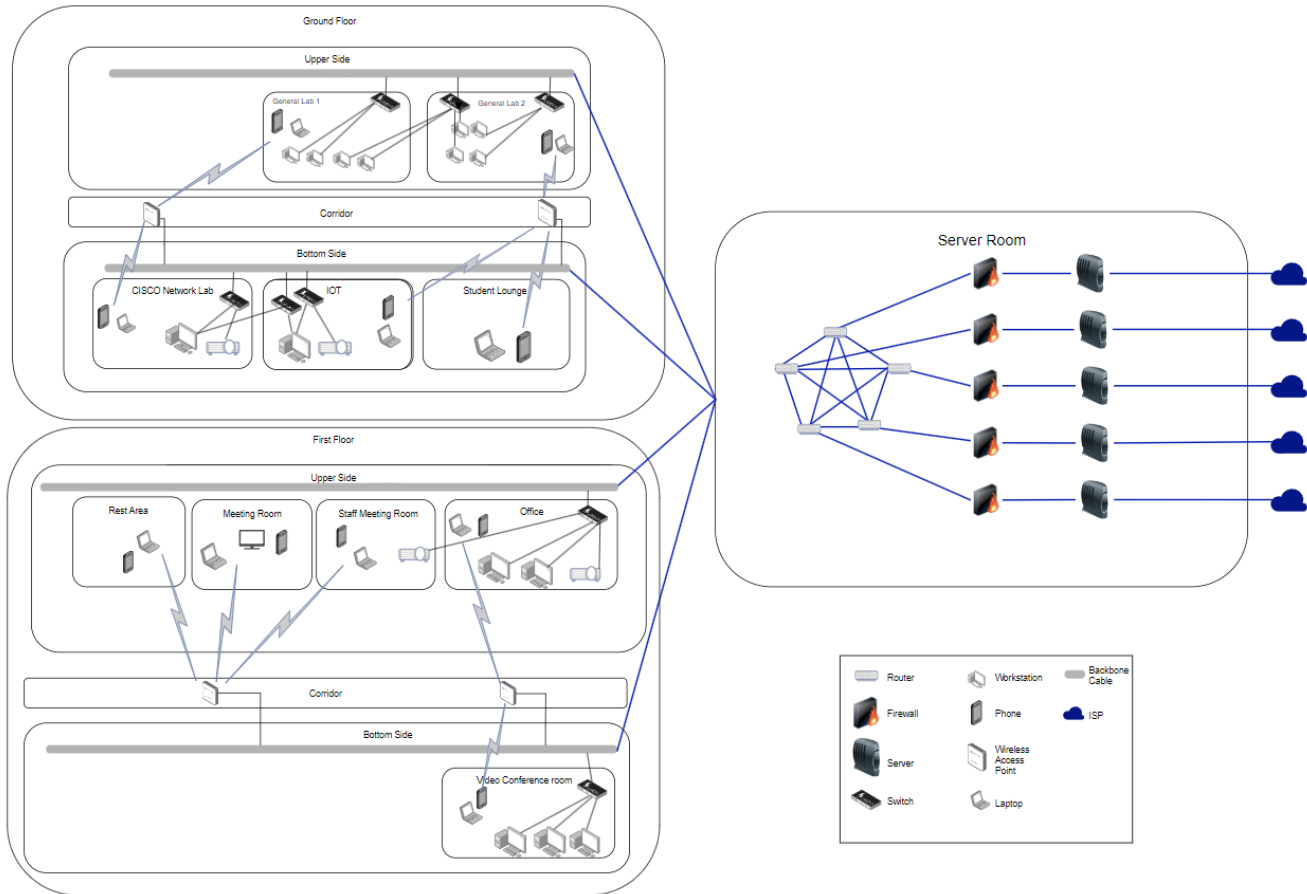


Figure 4.2.1.1: Network Distribution Diagram

The Network Distribution diagram illustrates the arrangement of network devices across different rooms, specifically focusing on how internet connectivity is established between them. Figure 4.2.1.1 shows the Network Distribution Diagram for the first floor and ground floor which consist of the upper side and lower side for each floor. Additionally, every room, with the exception of the rest area, meeting room, student lounge, and staff meeting room, contains a switch.

These areas like the rest area, meeting room, student lounge, and staff meeting room are connected to the wireless access points in the corridor directly connected to the backbone cable. This setup forms subnets, facilitating efficient device communication within each subnet without the need to pass through the router. This configuration aims to minimize latency, enhance transmission speed,

and reduce the router's workload.

For a clearer depiction of the network distribution on the second and first floors, refer to figures 4.2.1.2 and 4.2.1.3. It's important to note that the lines connecting each section of the building in the diagram do not precisely link to the router in the server room, a deliberate choice made to avoid visual confusion. Additional details explaining this aspect are provided in figure 4.2.4.

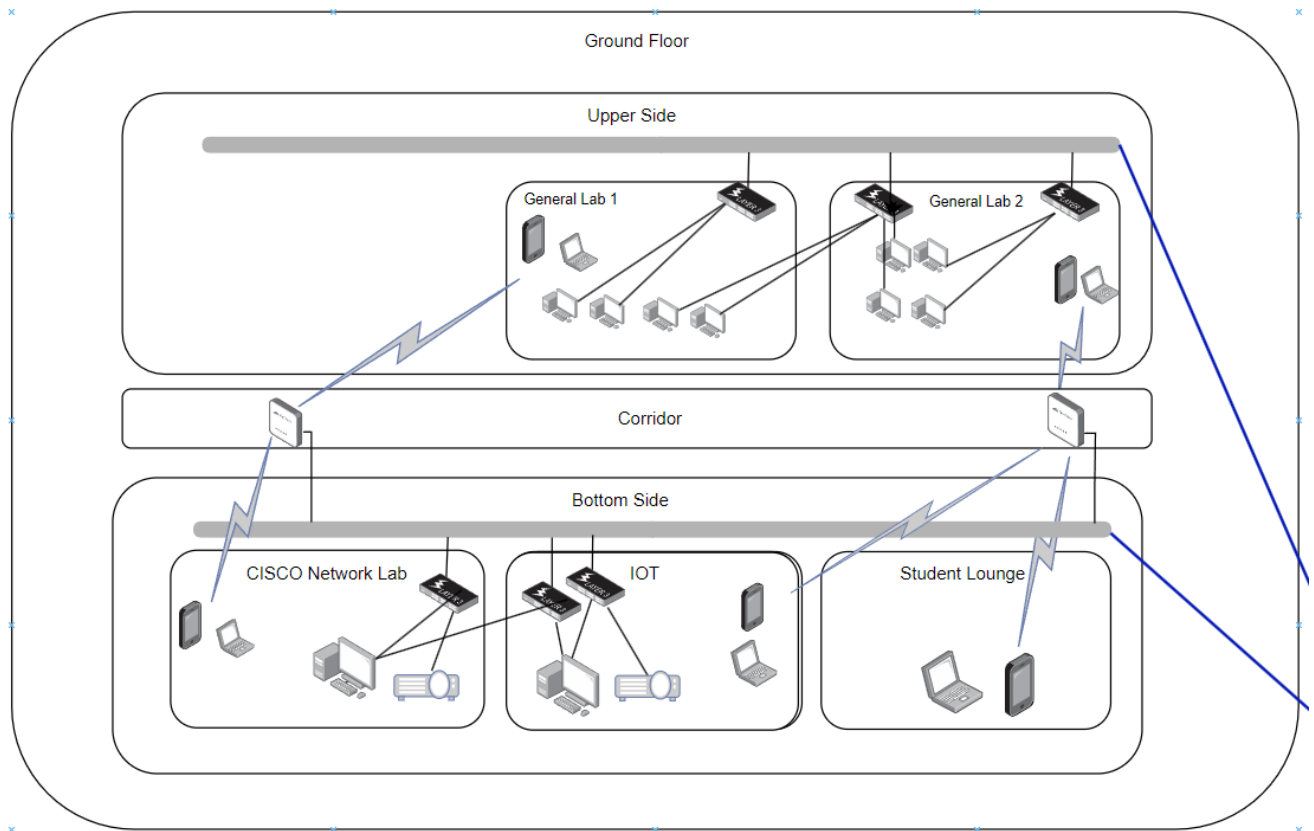


Figure 4.2.1.2: Network Distribution Diagram - Ground Floor

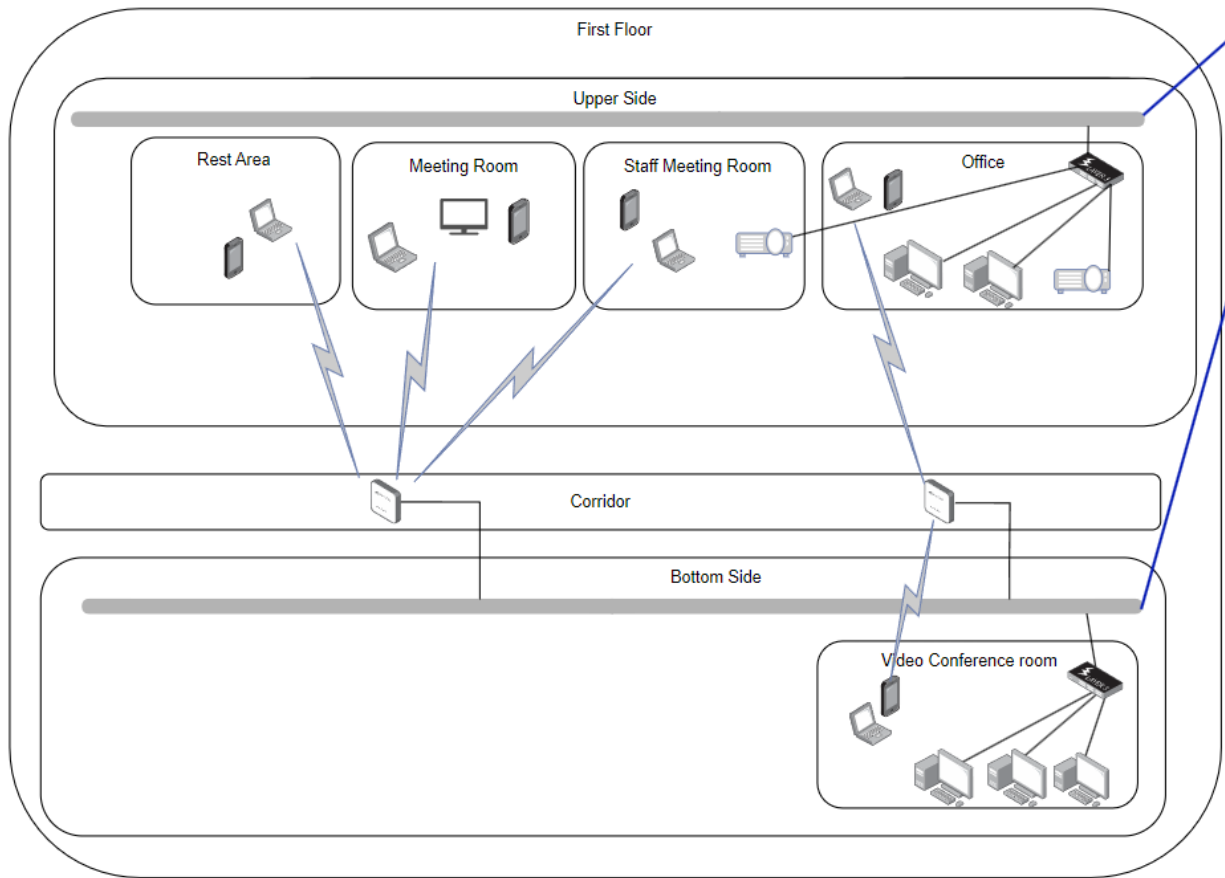


Figure 4.2.1.3: Network Distribution Diagram - First Floor

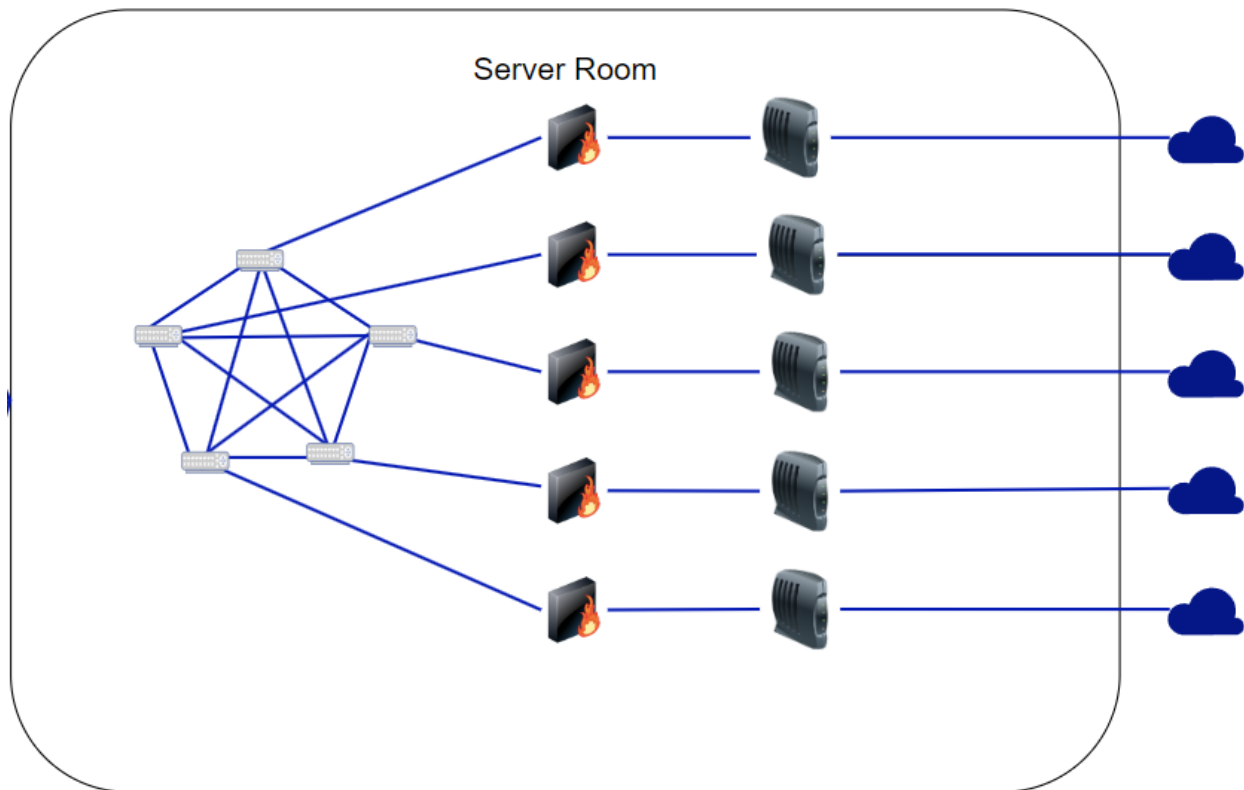


Figure 4.2.1.4: Network Distribution Diagram - Server Room

The depicted diagram outlines the network distribution within the server room, specifically focusing on the router configuration. In this setup, we plan to integrate five routers into the network, each connected independently through fiber optics. This design aims to prevent a single router failure from causing a complete network outage.

To further enhance resilience, the routers are interconnected using a mesh topology. This approach ensures that even if one router malfunctions, the network can continue to function. Additionally, in most rooms, a single switch is deployed to establish subnets, promoting subnet stability. Notably, one switch is redundantly connected to two routers, providing a failover mechanism to address potential emergencies or network disruptions.

4.2.2 Room Level Network Distribution Diagram

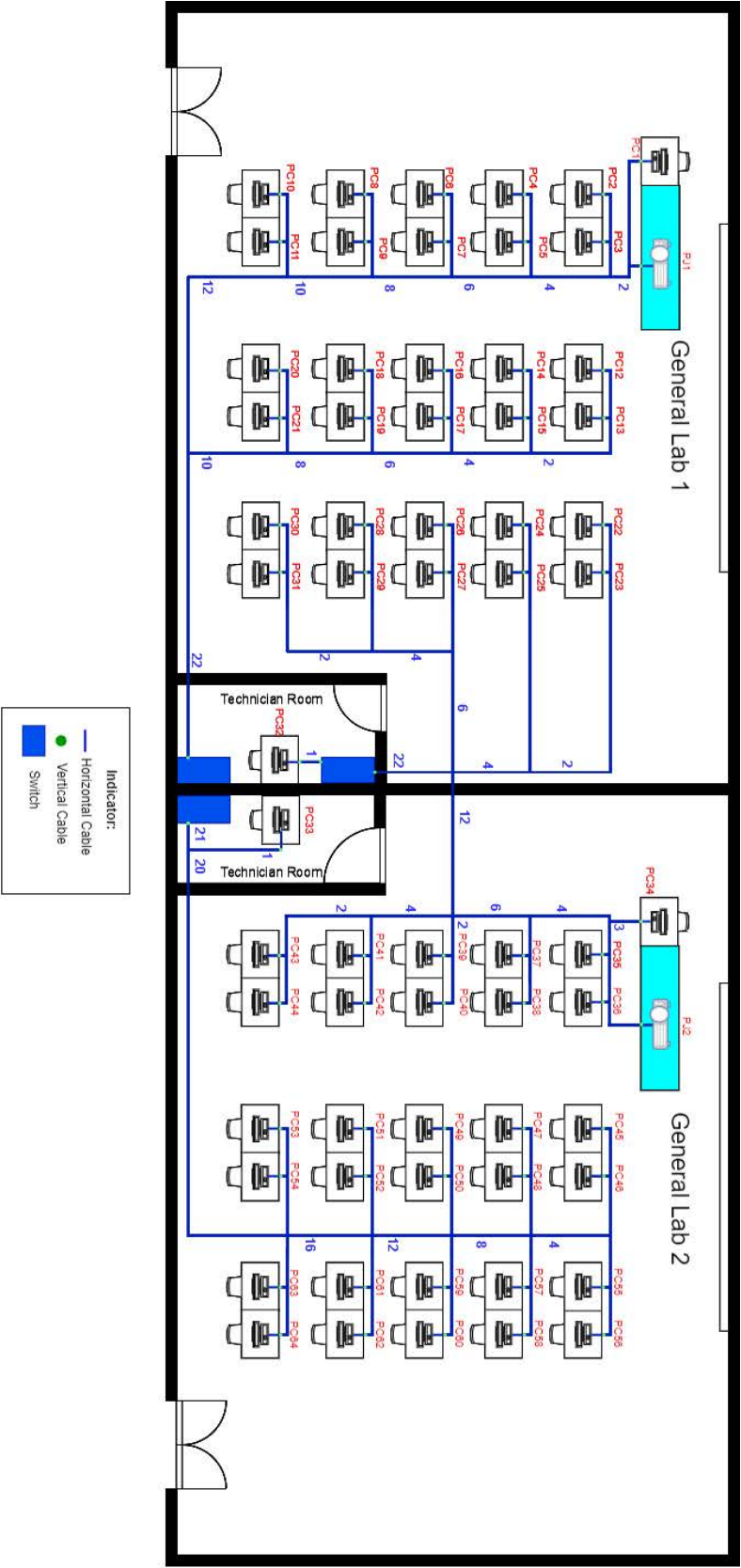


Figure 4.2.2.1: General Lab

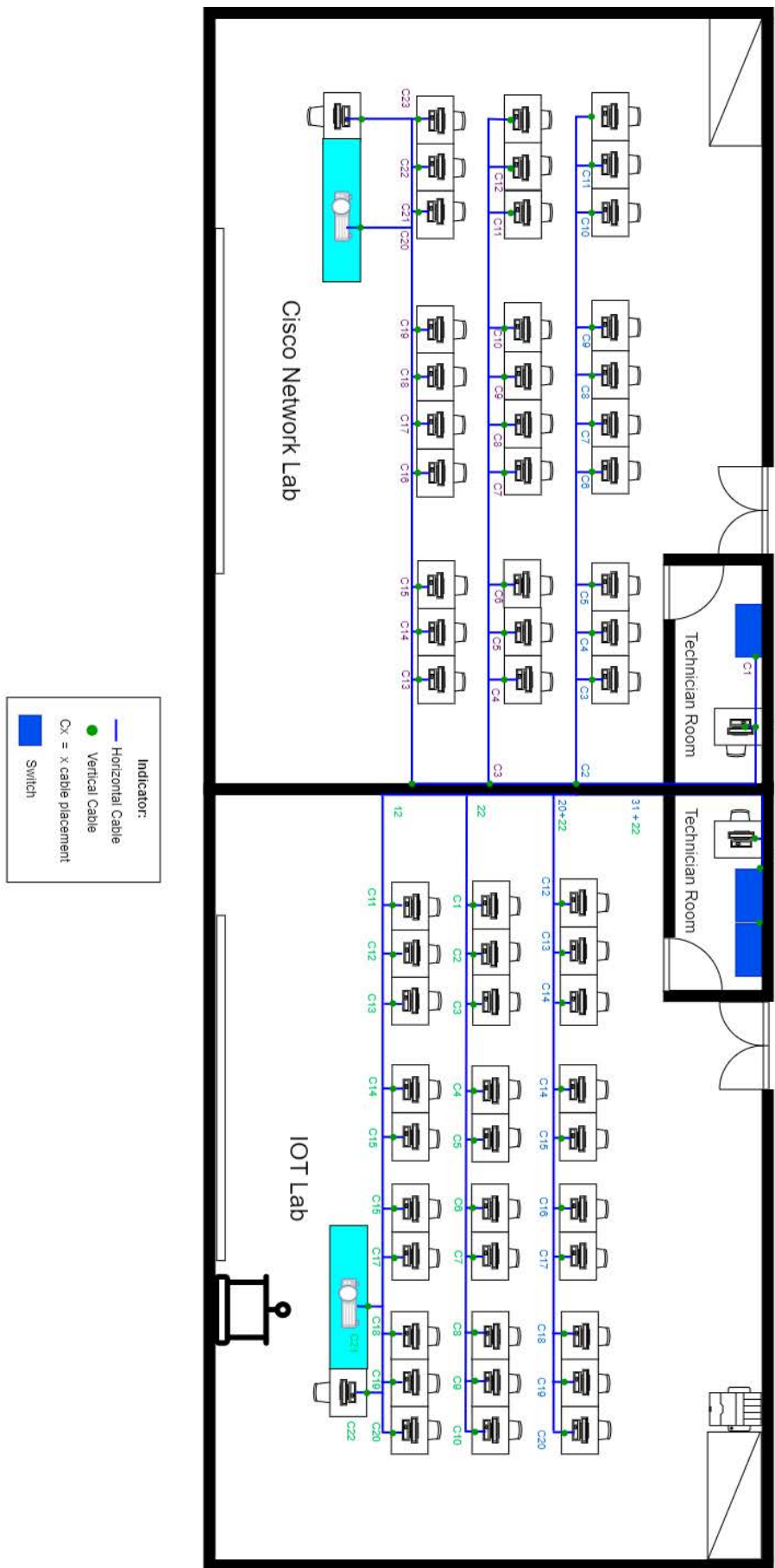


Figure 4.2.2.2: Cisco Network Lab and IoT Lab

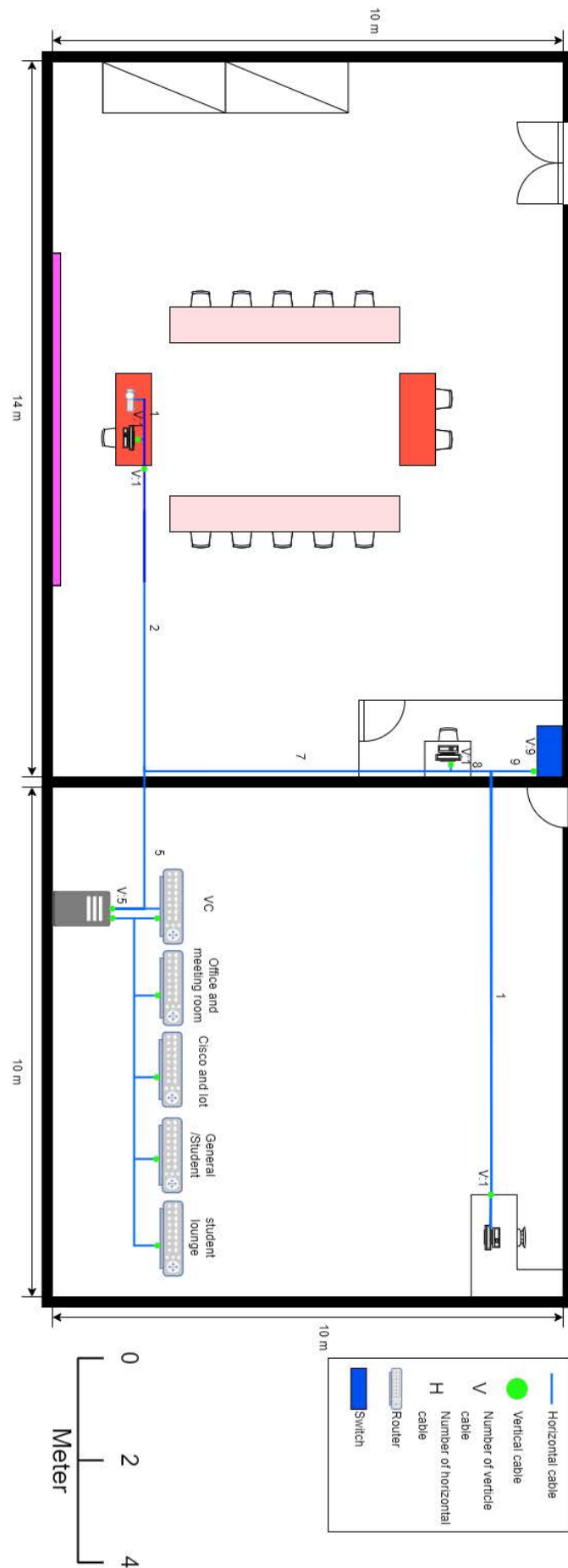


Figure 4.2.2.3: Video Conference Room and Server Room

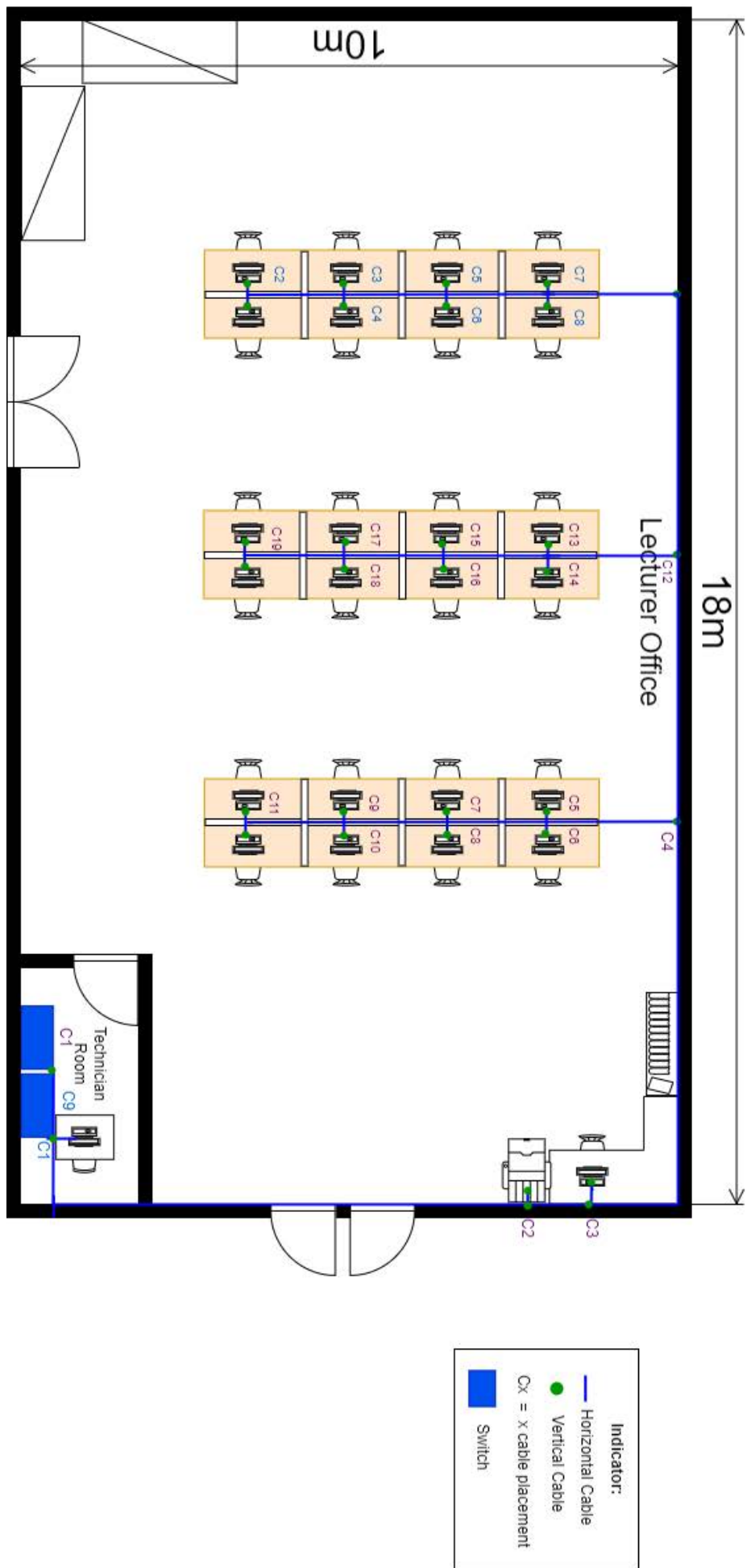


Figure 4.2.2.4: Lecturer Office

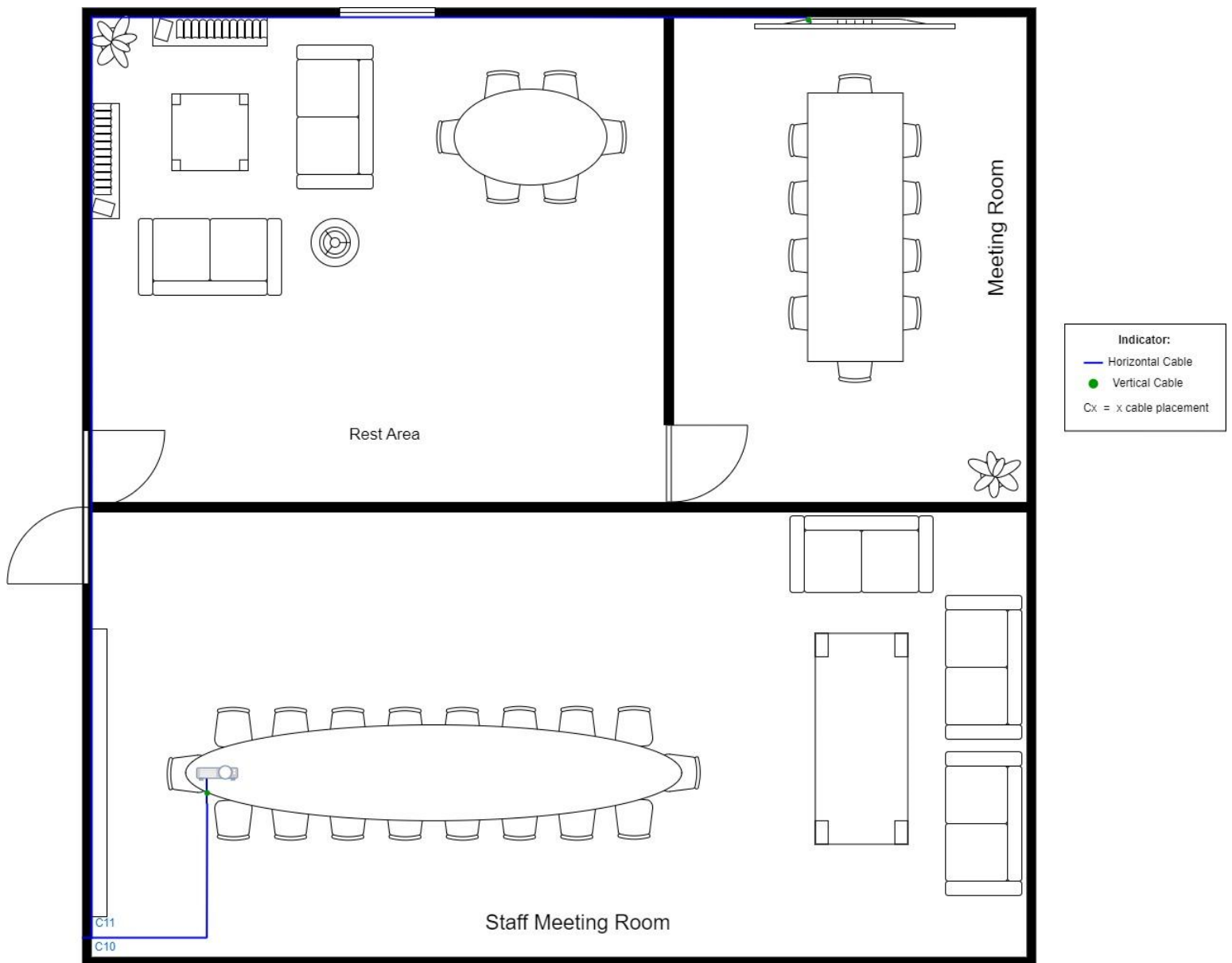


Figure 4.2.2.5: Meeting Room

4.2.3 Floor Diagram

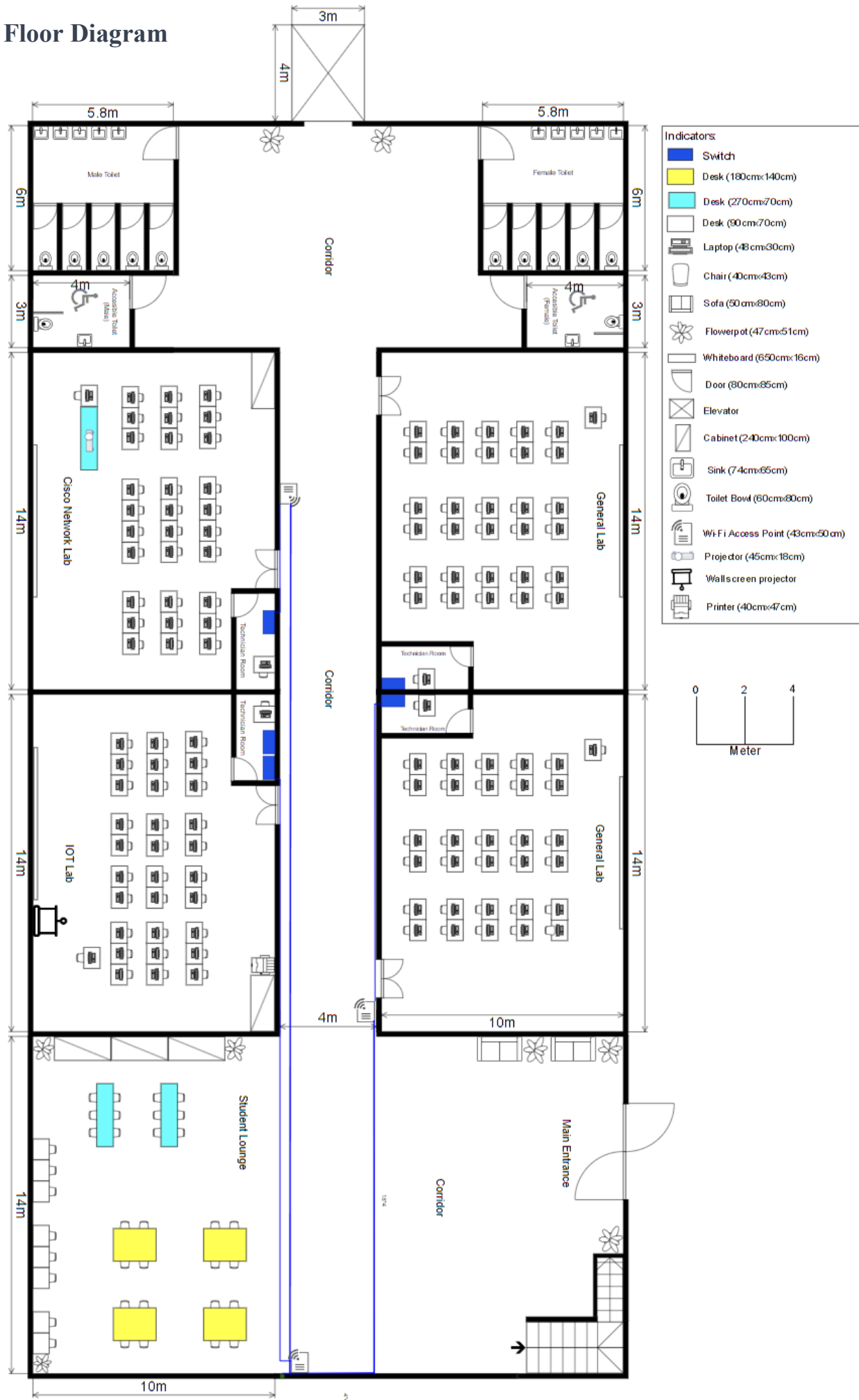
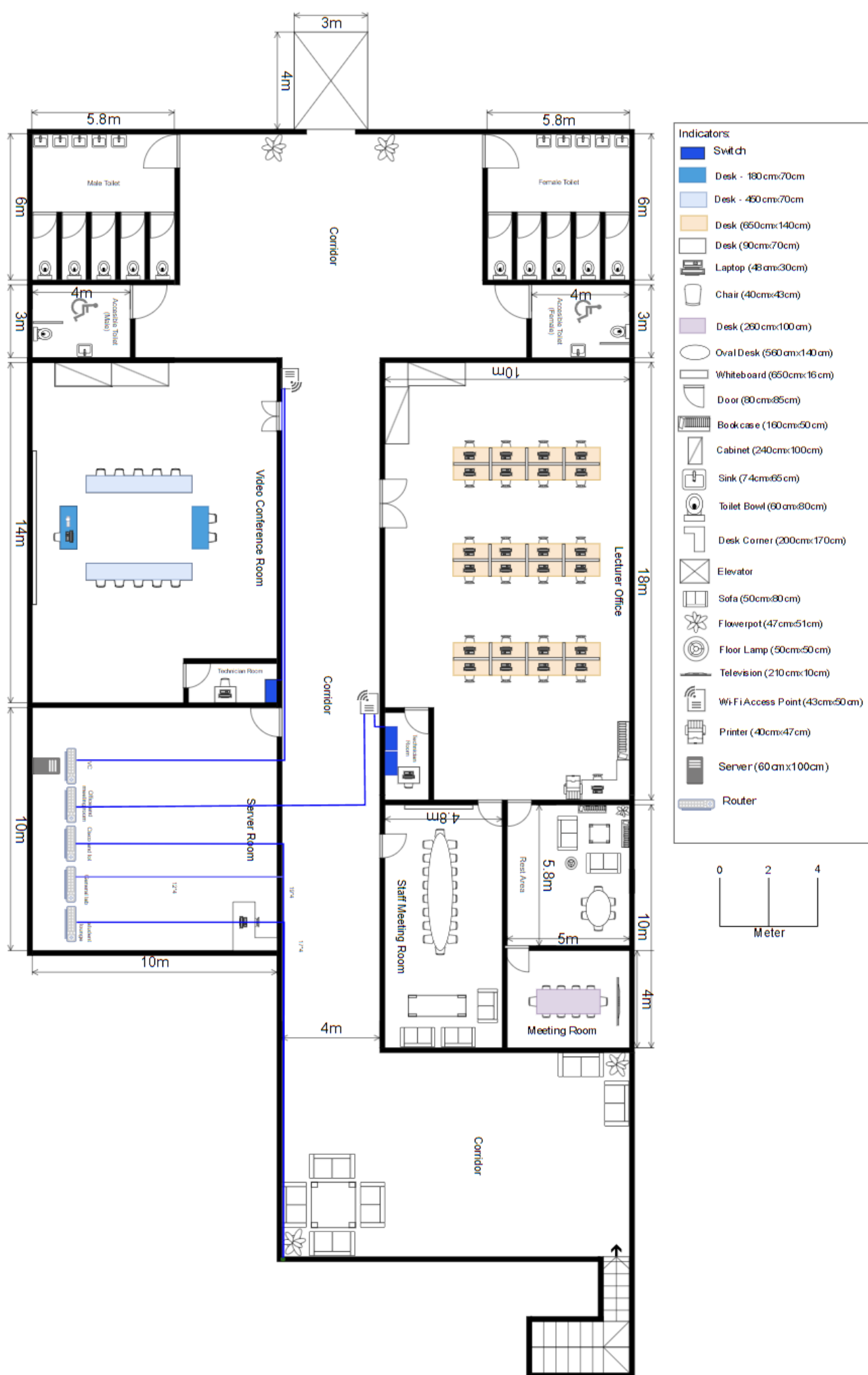


Figure 4.2.3.1: Cable Distribution of Ground Floor



The two floor diagrams as shown in Figure 4.2.3.1 and Figure 4.2.3.2 illustrate the planned network connections between individual rooms. All of the connections will originate from the server room where the router is situated. Each room will be linked to the router via a backbone connection, serving as a direct connection to establish a primary and robust connection for data transmission within the network.. Additionally, the diagram can depict the interconnection of wireless access points within the network in the ground and first floor of the faculty building.

4.3 Number of connections, patch cords and switch ports needed

Area	Number of Switch Ports Needed
General Lab 1	32
General Lab 2	32
Cisco Network Lab	33
IoT Lab	33
Video Conference Room	3
Office	27
Meeting Room	1
Staff Meeting Room	1
Server Room	6
Total	168

Table 4.3.1: Number of Switch Ports Needed

Area	Number of Patch Cords Needed
General Lab 1	32
General Lab 2	32
Cisco Network Lab	33
IoT Lab	33
Video Conference Room	3
Office	27
Meeting Room	1
Staff Meeting Room	1
Server Room	6
Total	168

Table 4.3.2: Number of Patch Cords Needed

Area	Number RJ45 Connector Needed
General Lab 1	64
General Lab 2	64
Cisco Network Lab	66
IoT Lab	66
Video Conference Room	6
Office	54
Meeting Room	2
Staff Meeting Room	2
Server Room	12
Total	335

Table 4.3.3: Number of RJ45 Connector Needed

4.4 Identify cable types and length

Area	Length (m)		Total Length (m)	Total Price of Cable Used (RM)
	Horizontal	Vertical		
Ground Floor				
Room				
General Lab 1	385.6	77.4	462	4325.94
General Lab 2	345.3	41.4	386.7	3620.87
Cisco Network Lab	341.8	11.8	353.6	3310.94
IOT Lab	649.4	91.3	740.7	6935.56
Ground Floor Room Total			1944.8	18210.17
First Floor				
Room				
Video Conference Room	31.7	5.8	37.5	351.13
Server Room	84	17.5	101.5	950.40
Office	600.8	18.8	619.6	5801.64
Staff Meeting Room	7	0.7	7.7	72.10
Meeting Room	35.4	0.7	36.1	338.02
First Floor Room Total			802.4	7513.29

Ground Floor		
Path	Length(m)	Price of cables used (RM)
Server room to General Lab 1 and 2	44	53.24
Server room to Cisco Network Lab	56.4	68.24
Server room to Student Lounge	28.8	34.85
Ground Floor Room Total	139.2	168.43
Total - Ground Floor		
Total length of wire to connect the room (m)	139.2	168.43
Total length of wire in different type (m)	1944.8	18210.17
Total length of wire needed (m)	2054.7	18378.60
First Floor		
Server Room to Video Conference Room	25.2	30.49
Server Room to Lecturer Office	20	24.20
First Floor Room Total	45.2	54.69
Total First Floor		
Total length of wire to connect the room (m)	45.2	54.69
Total length of wire in different type (m)	802.4	7513.29
Total length of wire needed (m)	847.8	7567.98

Total (Ground Floor + First Floor) (m)	2931.8	25946.58
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In the preceding section, we elucidated the distribution strategy and device connections. To optimize connectivity, careful consideration of cable types is crucial. Following our group deliberation, we've opted for a combination of CAT6 twisted cables and fiber optic cables within the building.

CAT6 twisted cables are selected for both horizontal and vertical computer cabling within rooms like the lab and lecture hall. This choice is driven by CAT6's ability to support high-speed data transmission, reaching up to 10 gigabits, owing to its impressive bandwidth of 550 MHz per cable. The thicker copper wire in CAT6 facilitates efficient heat dissipation, making it apt for Power over Ethernet (PoE), thereby reducing cabling time and costs. CAT6 also excels in supporting Ethernet applications across the building. The specific type of CAT6 chosen is Unshielded Twisted Pair (UTP), known for minimizing crosstalk and reducing electromagnetic interference caused by natural or human-made sources.

Additionally, fiber optic cables are designated for interconnecting routers in the server room and extending the connection to the Internet Services Provider (ISP). Leveraging light for data transmission, fiber optic cables ensure high-speed data transfer while mitigating signal degradation. With a bandwidth capability of up to 10Gbps, fiber optic cables are well-suited for handling substantial data volumes. Moreover, they excel in carrying signals across longer distances, ensuring fast and reliable data transmission.

In summary, our building necessitates an estimated length of cable, encompassing both CAT6 and fiber optic cables. This ensures seamless internal connectivity among devices within the building and robust linkage to the ISP.

Meeting Minutes #1

Date/Time		9 DEC 2023 8.00pm	
Location		Online (Google Meet)	
Agenda		1. Understand the details about task 4.	
		2. Discuss about the network distribution in our floor plan.	
		3. Standardize the necessary details such as height of the workstation from ground.	
		4. Task distribution for all members	
Meeting Host		Tan Jun Yuan	
Attendance			
Name		Time	Reason For Absence
Yong Wern Jie		2000	-
Chen Pyng Haw		2000	-
Koh Li Hui		2000	-
Tan Jun Yuan		2000	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1	Discussion for task 4	- Jun Yuan shared the information about task 4 and its rubric.	Jun Yuan
		- All members read and investigate the task carefully.	
		- Solved the problem raised by group members together through discussion.	
2	Suggestion for network distribution	- Jun Yuan and Wern Jie suggest the network	

		distribution in the floor plan.	All members
		- Pyng Haw and Li Hui suggest the necessary details that need to be standardized.	
3	Task distribution	- Wern Jie distributed the task to all members equally. - Pyng Haw was assigned to plan for the network distribution of IoT Lab. - Li Hui was assigned to plan for the network distribution of Cisco Network Lab. - Jun Yuan was assigned to plan for the network distribution of the video conference room. - Wern Jie was assigned to plan for network distribution of the general lab.	All members
4	Meeting Ended	- At 2230, the meeting ended after all discussions were done.	All members

Meeting Minutes #2

Date/Time		15 DEC 2023 11.30am	
Location		Online (Google Meet)	
Agenda		1. Review the current progress for each group member.	
		2. Finalize room level network distribution diagram	
		3. Finalize network distribution diagram	
		4. Check the calculation of the cable length needed	
Meeting MC		Koh Li Hui	
Attendance			
Name		Time	Reason For Absence
Yong Wern Jie		1130	-
Chen Pyng Haw		1130	-
Koh Li Hui		1130	-
Tan Jun Yuan		1130	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1	Review the current progress for each group member	- Wern Jie shared his part.	All members
		- Pyng Haw shared his part.	
		- Jun Yuan shared his part.	
		- Li Hui shared her part.	
2	Finalize network distribution diagram	- All members shared their part using drawio.	All members
3	Finalize room level network distribution diagram	- Li Hui suggested making the wiring more visible by adding colour and thickening it.	All members
		- Pyng Haw suggested labeling	

		the wiring.	
		- Wern Jie suggested the height of the switch and tables.	
		- Jun Yuan suggested to differentiate vertical and horizontal cable by using different labeling.	
		- All members share their ideas on how to improve the room level distribution diagram.	
4	Check the calculation of the cable length needed	- Li Hui reminded the other members about the correct calculation of cable length needed.	Li Hui
		- All members recheck the length calculation to make sure the accurate result will be obtained.	
5	Meeting ended	- At 0130, the meeting ended after all discussions were done.	All members